

# User Guide: CDP AI/ML Platform - Production ML Governance

This guide explains how to use the current CDP AI/ML Platform application for business review, model decisioning, live simulation, evidence inspection, and architecture review.

## 1. What The App Does

The application helps teams move from governed customer data to accountable AI/ML decisions. It is organized around four model workflows:

- TensorFlow Next Best Action Model
- TensorFlow Churn Propensity Model
- Bayesian Media Mix Optimization
- Causal Incrementality Model

Each workflow connects model inputs, recommendations, impact forecasts, decision evidence, activation outputs, and governance evidence.

## 2. Current App Navigation

The left navigation starts with the context page, then moves into the working pages:

| Order | Page | URL | Purpose |
|-------|------|-----|---------|
|-------|------|-----|---------|

|   |       |                           |                                                                                                             |
|---|-------|---------------------------|-------------------------------------------------------------------------------------------------------------|
| 1 | About | /ui/experience/about.html | Platform context, problem statement, use cases, governance model, production context, and architecture tabs |
|---|-------|---------------------------|-------------------------------------------------------------------------------------------------------------|

|   |                       |                                   |                                                            |
|---|-----------------------|-----------------------------------|------------------------------------------------------------|
| 2 | AI Decision Portfolio | /ui/experience/business-home.html | Business-facing model portfolio and recommended story flow |
|---|-----------------------|-----------------------------------|------------------------------------------------------------|

|   |                   |                                           |                                                     |
|---|-------------------|-------------------------------------------|-----------------------------------------------------|
| 3 | AI Impact Summary | /ui/experience/business-exec-summary.html | Executive portfolio health and model impact summary |
|---|-------------------|-------------------------------------------|-----------------------------------------------------|

|   |                          |                               |                                                      |
|---|--------------------------|-------------------------------|------------------------------------------------------|
| 4 | Model Decision Workbench | /ui/experience/workbench.html | Model-level decision walkthrough and evidence review |
|---|--------------------------|-------------------------------|------------------------------------------------------|

|   |                 |                                      |                                                                      |
|---|-----------------|--------------------------------------|----------------------------------------------------------------------|
| 5 | Simulation Flow | /ui/experience/index.html#simulation | Main action page for scenario selection and stage-by-stage execution |
|---|-----------------|--------------------------------------|----------------------------------------------------------------------|

|   |                     |                                        |                       |
|---|---------------------|----------------------------------------|-----------------------|
| 6 | Architecture Review | /ui/experience/index.html#architecture | Architecture path and |
|---|---------------------|----------------------------------------|-----------------------|

### 3. Recommended Walkthrough

Use this sequence for a complete demo:

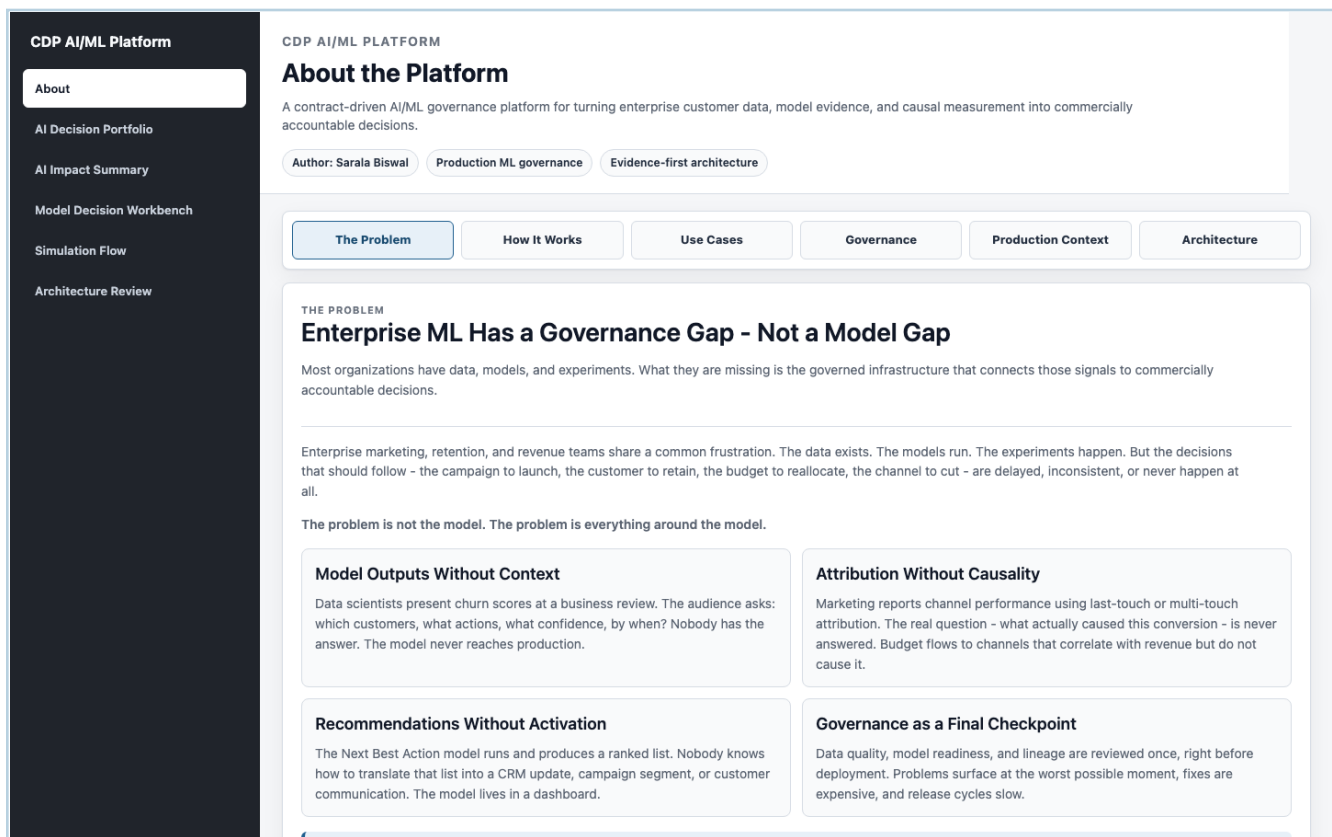
- Start with About to explain the governance gap and the platform thesis.
- Open AI Decision Portfolio to introduce the four model workflows.
- Open Model Decision Workbench for one model and walk from Model Inputs to Decision Evidence.
- Open AI Impact Summary for portfolio-level health and executive reporting.
- Open Simulation Flow to run the platform stage by stage.
- Use Run Evidence and Architecture Review to explain artifacts, lineage, and governance.

### 4. Visual Walkthrough

#### Step 1: Start With About

Use About to explain why the platform exists before showing the operational screens. The page has six tabs: The Problem, How It Works, Use Cases, Governance, Production Context, and Architecture.

## About



What to point out:

- The platform is focused on the governance gap around enterprise ML, not only model training.
- The six tabs separate business context, implementation model, use cases, governance, production history, and architecture.
- About is the first navigation item across the app.

## Step 2: Open AI Decision Portfolio

Use AI Decision Portfolio to introduce the model portfolio and the recommended presentation sequence.

## AI Decision Portfolio

The screenshot displays the 'CDP AI/ML Platform' interface. On the left is a dark sidebar with navigation links: 'About', 'AI Decision Portfolio' (highlighted), 'AI Impact Summary', 'Model Decision Workbench', 'Simulation Flow', and 'Architecture Review'. The main content area is titled 'CDP AI/ML PLATFORM AI Decision Portfolio' and includes a subtitle: 'A governed portfolio of AI/ML decision models for customer growth, retention, marketing optimization, and causal measurement, with evidence from model input to activation.' Below this are three filters: 'AI Portfolio: Growth + Retention', 'Model Cadence: Weekly / Monthly', and 'Audience: Business + AI Leaders'. A section titled 'How To Present The AI/ML Story' provides a sequence of steps: 1. AI Decision Portfolio, 2. AI Impact Summary, 3. Model Decision Workbench, 4. Simulation Flow, and 5. Architecture Review. It includes buttons for 'Open AI Impact Summary' and 'Open Model Workbench'. The main area features four model cards: 1. 'TensorFlow Next Best Action Model' with 'MODEL EVIDENCE: LATEST RUN', describing a model for customer actions and channels, with a button 'Open Model Workflow'. 2. 'TensorFlow Churn Propensity Model' with 'MODEL EVIDENCE: LATEST RUN', predicting churn risk and retention impact, with a button 'Open Model Workflow'. 3. 'Bayesian Media Mix Optimization' with 'MODEL EVIDENCE: LATEST RUN', using Bayesian marketing science for response curves, with a button 'Open Model Workflow'. 4. 'Causal Incrementality Model' with 'MODEL EVIDENCE: LATEST RUN', separating true campaign lift from baseline demand, with a button 'Open Model Workflow'.

What to point out:

- Four governed AI/ML decision workflows.
- Each model has a decision output and primary business metric.
- Each card opens the Model Decision Workbench for that scenario.

### Step 3: Review Model Inputs

Open a model workflow and start with Model Inputs. This step shows whether governed source volume, features, and scorable rows are ready before recommendations are reviewed.

## Model Inputs

The screenshot displays the 'Model Decision Workbench' interface within the 'CDP AI/ML Platform'. The interface is divided into three main sections: a left sidebar, a central workspace, and a right sidebar.

**Left Sidebar:** Contains navigation links for 'About', 'AI Decision Portfolio', 'AI Impact Summary', 'Model Decision Workbench' (highlighted), 'Simulation Flow', and 'Architecture Review'.

**Central Workspace:** Titled 'Model Decision Workbench', it includes a subtitle 'Inspect how AI/ML models convert governed inputs into recommendations, impact forecasts, decision evidence, and activation-ready outputs.' Below this, there are two main panels:

- AI MODEL:** A list of models including 'TensorFlow NBA' (selected), 'TensorFlow Churn', 'Bayesian MMM', and 'Causal Lift'.
- EVIDENCE VIEW:** A list of evidence types including 'Model Decision' (selected), 'Model Evidence', and 'Model Input'.
- STEP:** A list of steps including '1. Model Inputs' (selected), '2. Model Recommendation', '3. Impact Forecast', and '4. Decision Evidence'.

**Model Decision Panel:** Displays the 'TensorFlow Next Best Action Model' with a 'Refresh evidence' button. It shows 'Step 1: Model Inputs' and a narrative: 'AI decision narrative with model inputs, recommendations, forecasts, and evidence in sequence.' Below this is a 'Model Input Readiness' section with three metrics:

| SOURCE TABLES             | FEATURE ROWS             | MODEL ROWS          |
|---------------------------|--------------------------|---------------------|
| 3                         | 100                      | 100                 |
| Contract inputs available | Prepared for model stage | Scorable population |

Below the metrics, there are three text boxes:

- Top observed reason: high\_churn\_risk (100 rows).
- 100 feature rows prepared for scoring.
- 100 rows entered model stage in latest run.

Navigation buttons 'Previous Step' and 'Next Step' are at the bottom.

**Right Sidebar:** Contains a 'Model Run Snapshot' section with details: Run ID (20260527T014835Z), Status (PASS), Stage Pass (100%), and Runtime (synthetic\_only). Below this is a 'Quick Actions' section with buttons for 'Open Model Workflow' and 'Open AI Impact Summary'.

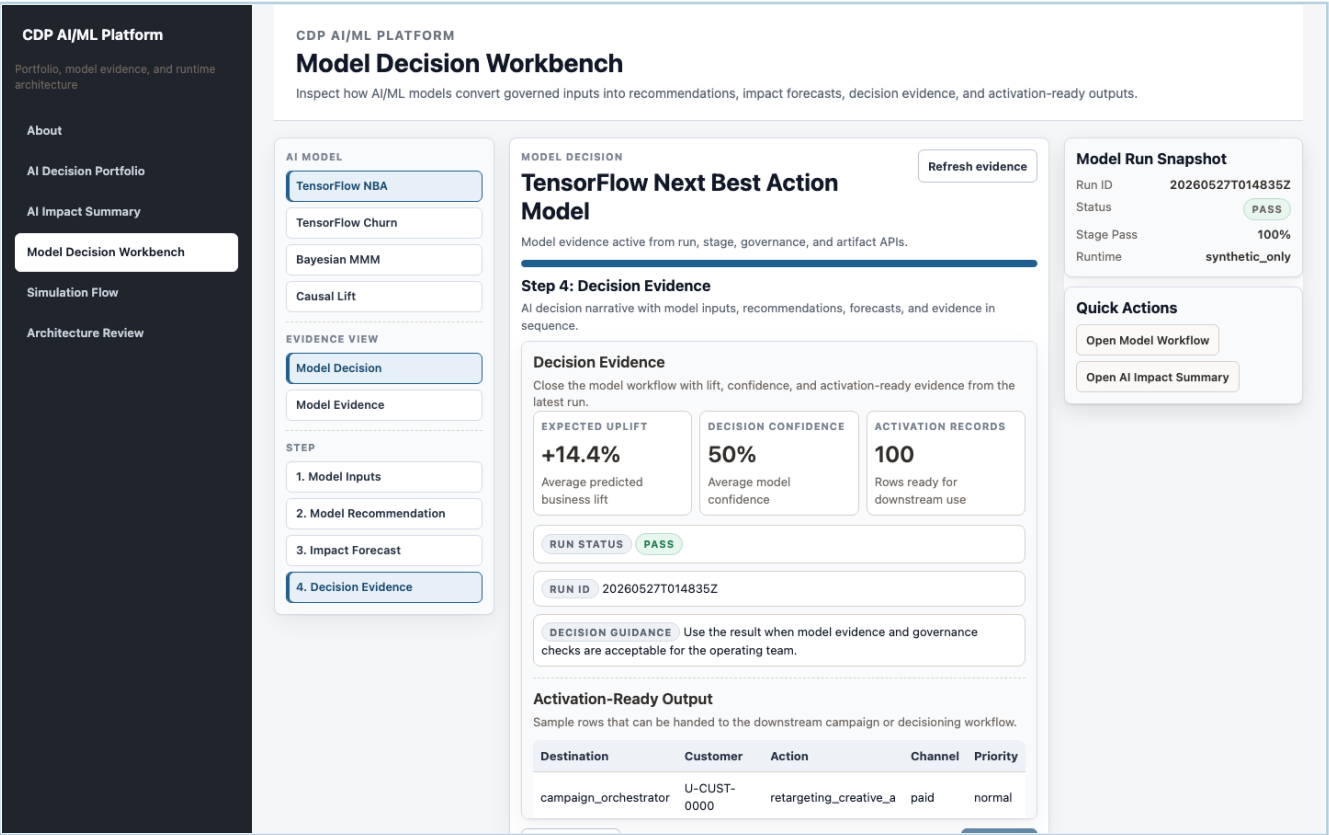
What to point out:

- The left rail keeps model selection, evidence view, and step navigation visible.
- The center panel explains one decision step at a time.
- The right panel confirms run ID, status, stage pass rate, and runtime.

## Step 4: Review Decision Evidence

Move to Decision Evidence to close the business workflow with uplift, confidence, activation records, and run status.

# Decision Evidence



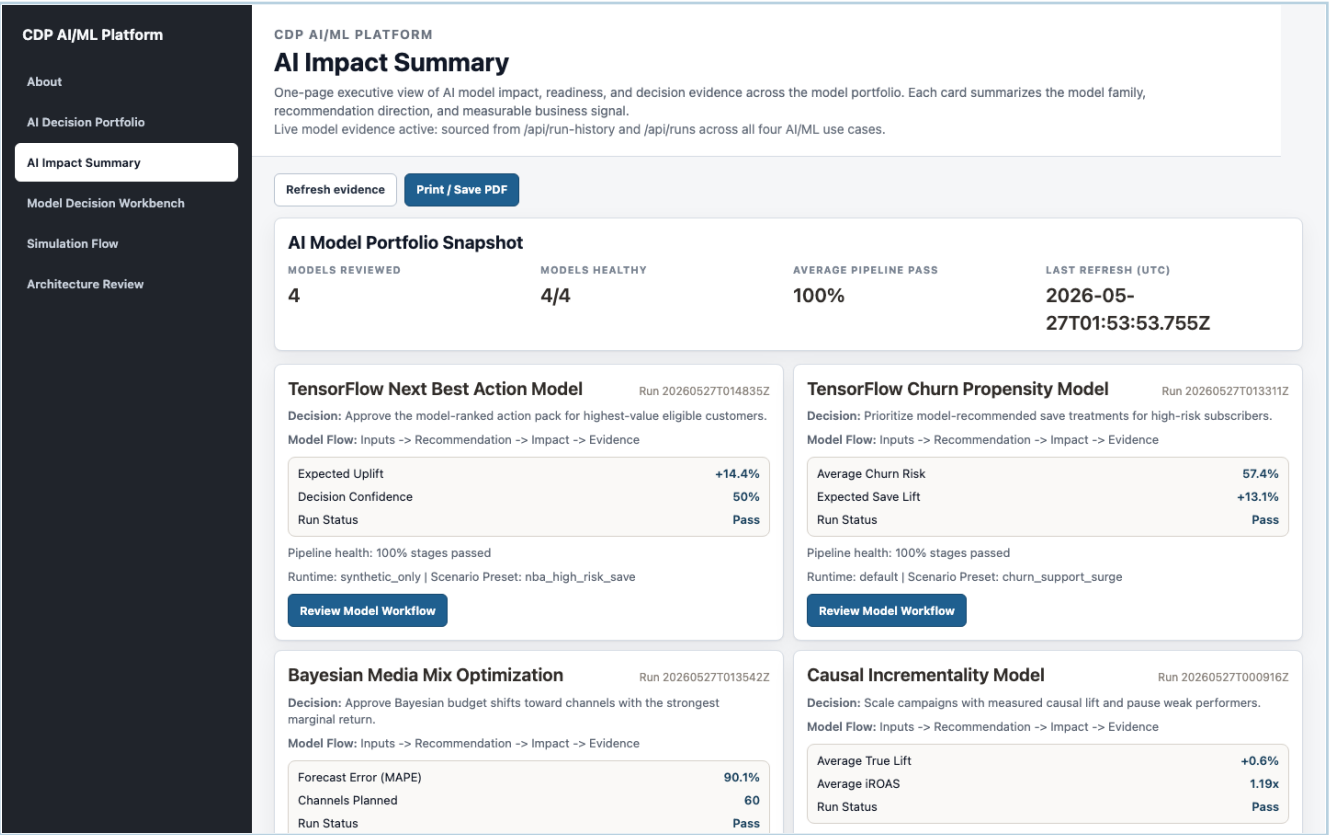
What to point out:

- Expected uplift summarizes predicted business value.
- Decision confidence gives the average model confidence signal.
- Activation records show that outputs are ready for downstream decisioning.

## Step 5: Review AI Impact Summary

Open AI Impact Summary for an executive view across the model portfolio.

# AI Impact Summary



What to point out:

- Portfolio-level model health and latest run status.
- Model-level decisions, KPIs, and readiness.
- The page can be printed or saved as a PDF for executive review.

## Step 6: Run Simulation Flow

Simulation Flow is the main action page. It is intentionally focused on the controls used to run or explain the pipeline.

# Simulation Flow

CDP AI/ML Platform

Production ML governance, model evidence, and activation readiness.

About

Overview

Model Portfolio

Run Evidence

Simulation

Architecture

MODELS IN SCOPE

4

LATEST RUN STATUS

Pass

PIPELINE STAGES

8

RUNTIME PROFILE

Local

SIMULATION

Run the pipeline stage by stage

Select a model workflow and scenario preset, then advance the governed pipeline with live evidence.

AI MODEL

TensorFlow Churn Pr

SCENARIO PRESET

none

RUNTIME MODE

synthetic\_only

INFRA PROFILE

local

Run next stage

Run all stages

Reset

STATUS

Ready

SESSION

Not started

PROGRESS

0 / 8

1 Data Sources

2 Ingestion + Event Bus

3 Raw + Curated Storage

4 Identity + Customer 360

5 Feature Layer

6 Model Layer

7 Serving + Activation

8 Monitoring + Governance

Latest run snapshot

Use case

UC-CHURN-RET-002

Run status

Pass

Infra profile

Local

Stage pass ratio

8 / 8

Model registry

Candidate

Evidence center

Stage telemetry

Open

Data quality gates

Open

Model metrics

Open

Activation payload

Open

Lineage manifest

Open

Architecture path

| LAYER      | PURPOSE                     |
|------------|-----------------------------|
| Data       | Source tables and readiness |
| Features   | Reusable model inputs       |
| Model      | Predictions and metrics     |
| Governance | Approval and monitoring     |

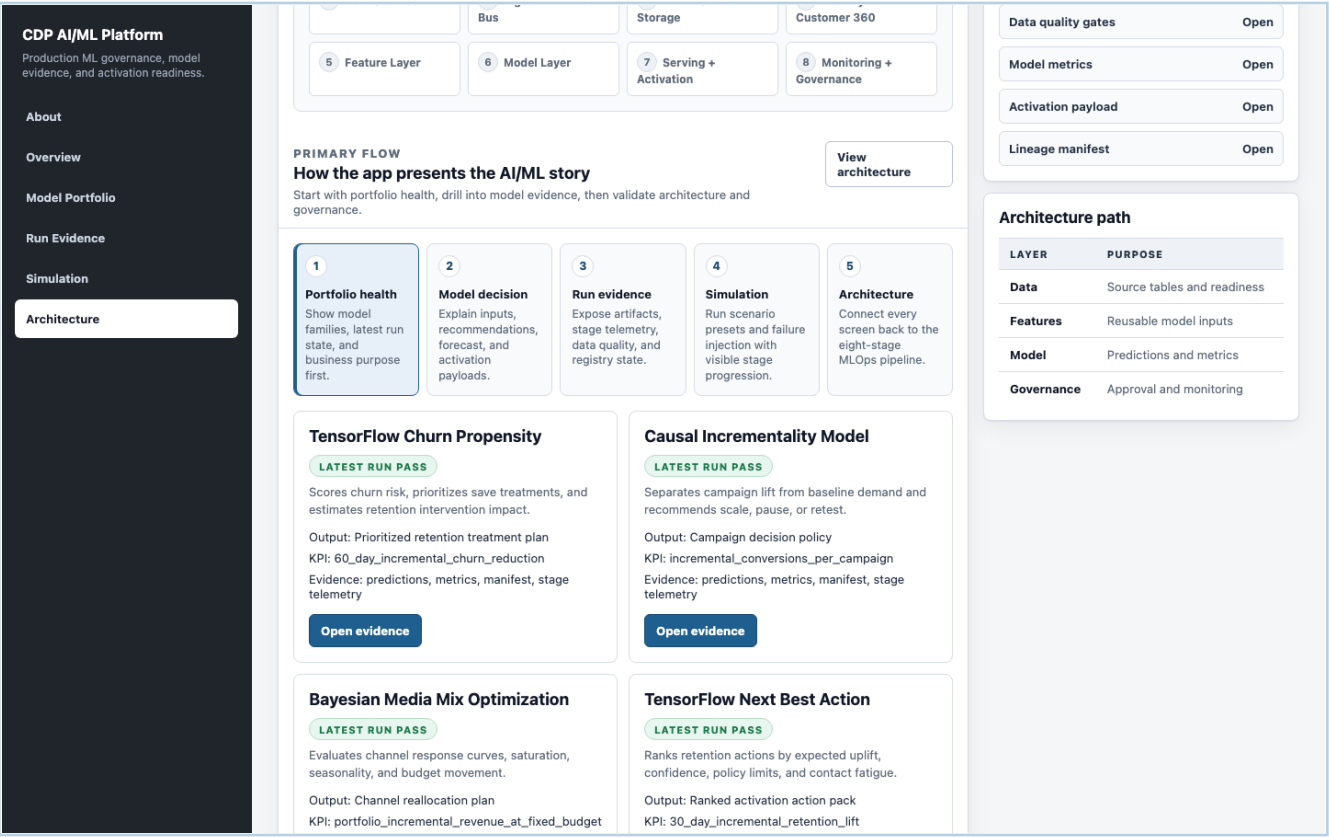
What to point out:

- Select an AI model, scenario preset, runtime mode, and infra profile.
- Use Run next stage to advance the pipeline gradually.
- Use Run all stages for a full scenario execution.
- The stage tiles show progress across the eight-stage governed pipeline.

## Step 7: Review Architecture

Use Architecture Review to connect the user experience back to the platform layers and governance path.

# Architecture Review



What to point out:

- Data, feature, model, and governance layers are shown as the operating path.
- Architecture is secondary to the action workflow, but useful for technical review.
- The same pipeline stages support all four model workflows.

## Step 8: Inspect Run Evidence

Use Run Evidence and the Evidence Center to open stage telemetry, data quality gates, model metrics, activation payloads, and lineage manifests.

# Run Evidence

CDP AI/ML Platform

Production ML governance, model evidence, and activation readiness.

About

Overview

Model Portfolio

Run Evidence

Simulation

Architecture

SIMULATION

Run the pipeline stage by stage

Select a model workflow and scenario preset, then advance the governed pipeline with live evidence.

AI MODEL

TensorFlow Churn Pt

SCENARIO PRESET

none

RUNTIME MODE

synthetic\_only

INFRA PROFILE

local

Run next stage

Run all stages

Reset

STATUS

Ready

SESSION

Not started

PROGRESS

0 / 8

1 Data Sources

2 Ingestion + Event Bus

3 Raw + Curated Storage

4 Identity + Customer 360

5 Feature Layer

6 Model Layer

7 Serving + Activation

8 Monitoring + Governance

PRIMARY FLOW

How the app presents the AI/ML story

Start with portfolio health, drill into model evidence, then validate architecture and governance.

1 Portfolio health

2 Model decision

3 Run evidence

4 Simulation

5 Architecture

TensorFlow Churn Propensity

LATEST RUN PASS

Scores churn risk, prioritizes save treatments, and estimates retention intervention impact.

Causal Incrementality Model

LATEST RUN PASS

Separates campaign lift from baseline demand and recommends scale, pause, or retest.

Latest run snapshot

Use case

UC-CHURN-RET-002

Run status

Pass

Infra profile

Local

Stage pass ratio

8 / 8

Model registry

Candidate

Evidence center

Stage telemetry

Open

Data quality gates

Open

Model metrics

Open

Activation payload

Open

Lineage manifest

Open

Architecture path

LAYER

PURPOSE

Data

Source tables and readiness

Features

Reusable model inputs

Model

Predictions and metrics

Governance

Approval and monitoring

What to point out:

- Evidence links are generated from the latest run artifacts.
- Stage telemetry and data quality are directly inspectable.
- Activation payload and lineage links close the audit trail.

## Step 9: Switch To Technical Model Evidence

In the Model Decision Workbench, switch from Model Decision to Model Evidence for a deeper technical review.

## Technical Model Evidence

The screenshot displays the CDP AI/ML Platform Model Decision Workbench. The interface is divided into several sections:

- Left Sidebar:** Contains navigation links for 'About', 'AI Decision Portfolio', 'AI Impact Summary', 'Model Decision Workbench' (highlighted), 'Simulation Flow', and 'Architecture Review'.
- Top Header:** Displays 'CDP AI/ML PLATFORM' and 'Model Decision Workbench'.
- AI MODEL Section:** Lists models: TensorFlow NBA (selected), TensorFlow Churn, Bayesian MMM, and Causal Lift.
- EVIDENCE VIEW Section:** Includes 'Model Decision' and 'Model Evidence' (selected).
- STEP Section:** Shows a sequence of steps: '1. Data + Model Contract' (selected), '2. Model Runtime + Activation', and '3. Responsible AI Evidence'.
- MODEL EVIDENCE Section:** Features a 'Refresh evidence' button and the title 'TensorFlow Next Best Action Model'. It provides a deep technical walkthrough of contracts, stage evidence, model lineage, and governance artifacts.
- Data Contract + Stage Inputs:** A detailed view of the data contract and stage inputs, including a table of contract controls and policy rules.
- Model Run Snapshot:** Displays run details: Run ID (20260527T014835Z), Status (PASS), Stage Pass (100%), and Runtime (synthetic\_only).
- Model Evidence Snapshot:** Shows components (8/8 pass), artifacts (17/28 present), data quality (not\_applicable (0/0)), and model backend (tensorflow).
- Quick Actions:** Includes buttons for 'Open Model Workflow' and 'Open AI Impact Summary'.

What to point out:

- Technical evidence connects components, APIs, stage outputs, and artifacts.
- The same workflow can be explained to business or technical audiences without changing apps.
- Model evidence supports readiness and governance conversations.

## 5. Starting The App

From the repository root:

```
make standalone
```

Open:

```
http://127.0.0.1:8080/ui/experience/about.html
```

If dependencies and artifacts already exist:

```
make standalone-fast
```

For live UI/API mode:

```
make ui-live
```

## 6. Business User Guide

Business users should focus on the decision story and model impact.

### Step 1: Open About

Start with the platform thesis:

- Enterprise ML has a governance gap, not a model gap.
- The platform connects data, models, evidence, and activation.
- The four use cases cover retention, churn, media mix, and incrementality.

### Step 2: Open AI Decision Portfolio

Use the portfolio page to understand which models are available and what each model is designed to decide.

Look for:

- Model name
- Decision output
- Primary business metric
- Recommended presentation sequence

### Step 3: Open A Model Workflow

Click Open Model Workflow for one of the models.

Recommended first workflow:

TensorFlow Next Best Action Model

## **Step 4: Walk The Business Steps**

In the Model Decision Workbench, use the left-side step navigation:

- Model Inputs
- Model Recommendation
- Impact Forecast
- Decision Evidence

Use these questions while presenting:

- What customer, campaign, or spend signals entered the model?
- What does the model recommend?
- What impact does the model forecast?
- What evidence supports the decision?

## **Step 5: Review AI Impact Summary**

Open AI Impact Summary to see portfolio-level model health.

Look for:

- Models reviewed
- Models healthy
- Average pipeline pass rate
- Latest run status
- Model-specific business KPIs

## 7. Technical User Guide

Technical users should focus on simulation, stage evidence, artifacts, and governance.

### Step 1: Open Simulation Flow

Use Simulation Flow as the operational surface.

Controls:

- AI model - selects the use case.
- Scenario preset - selects deterministic scenario behavior.
- Runtime mode - choose synthetic\_only or default.
- Infra profile - choose local or oss.
- Run next stage - advances one stage.
- Run all stages - executes the full eight-stage flow.
- Reset - resets the simulation session.

The pipeline stages are:

Data Sources

- > Ingestion + Event Bus
- > Raw Storage + Curated Warehouse
- > Identity Resolution + Customer 360
- > Feature Layer
- > Model Layer
- > Serving + Activation
- > Monitoring + Governance

### Step 2: Use Scenario Presets

Available deterministic presets include:

Use Case | Scenario Preset

UC-NBA-RET-001 | nba\_high\_risk\_save

UC-NBA-RET-001 | nba\_dq\_failure\_demo

UC-CHURN-RET-002 | churn\_support\_surge  
UC-MMM-PLN-003 | mmm\_budget\_rebalance  
UC-INCR-MKT-004 | incrementality\_negative\_lift

### Step 3: Inspect Run Evidence

Use Run Evidence or the right-side Evidence Center to inspect:

- Stage telemetry
- Data quality gates
- Model metrics
- Activation payload
- Lineage manifest

### Step 4: Use Model Evidence View

Open Model Decision Workbench and select Model Evidence.

Use this view to inspect:

- Contract controls
- Stage inputs and outputs
- Model backend
- Model metrics
- Activation payloads
- Validation gates
- Artifact paths

## 8. Runtime Profiles

Profile | Use When

Local / Standalone | You want the app to run locally without external services.

OSS Integrated | You want to show Kafka, Postgres, and MinIO-style infrastructure mirroring.

Enterprise Integration | You want to show readiness for MLflow, Feast, Splink, Airflow, Keycloak, monitoring, and metrics.

The app can run standalone without deployment dependencies. Enterprise integrations are optional.

## 9. Common Commands

### Run all model workflows

```
python3 -m pipelines.cli run-stack-all
```

### Run one workflow

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001 \
```

### Run a deterministic scenario

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001 \
  --runtime-mode synthetic_only \
  --scenario-id nba_high_risk_save
```

### Run a governance failure demo

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001 \
  --runtime-mode synthetic_only \
  --scenario-id nba_dq_failure_demo
```

### View recent runs

```
python3 -m pipelines.cli list-runs
```

### Inspect run stages

```
python3 -m pipelines.cli show-stages --use-case UC-NBA-RET-001 --run-id <run_id>
```

## Inspect model readiness

```
python3 -m pipelines.cli model-readiness --use-case UC-NBA-RET-001 --run-id <run_id>
```

## 10. Troubleshooting

If the app is not loading:

- Confirm the server is running.
- Confirm the URL uses the right port.
- Run make standalone-fast if dependencies and artifacts already exist.
- Run make smoke-standalone to validate the app.

If model evidence is missing:

- Run at least one workflow.
- Refresh the page.
- Check artifacts/<use\_case\_id>/<run\_id>/summary.json.

If Simulation controls do not respond:

- Confirm the live server is running with make ui-live or make standalone.
- Confirm /api/health returns ok.
- Use the local infra profile first, then switch to oss when optional services are available.

## 11. What To Show In A Meeting

For business stakeholders:

- About
- AI Decision Portfolio
- One Model Decision Workbench workflow

- AI Impact Summary

For technical stakeholders:

- About - Architecture tab
- Simulation Flow
- Run Evidence
- Model Evidence View
- Architecture Review

For platform leaders:

- Business value from model workflows
- Evidence-backed governance
- Simulation-driven execution
- Runtime profiles and enterprise integration path